Partial Fractions

We can express the sum or difference of two rational expressions $\frac{3}{x-1}$ and $\frac{2}{x-2}$ as a single rational expression.

i.e.,

$$\frac{3}{x-1}+\frac{2}{x-2} = \frac{5x-8}{(x-1)(x-2)}$$

$$\frac{5x-8}{(x-1)(x-2)} = \frac{3}{x-1}+\frac{2}{x-2}$$

Process of writing a single rational expression as a sum or difference of two or more simple rational expressions is called splitting up into *partial fractions*.

Generally if $p(x)$ and $q(x)$ are two rational integral algebraic functions of x and the fraction $\frac{p(x)}{q(x)}$ be expressed as the algebraic sum (or difference) of simpler fractions according to certain specified rules, then the rational expression $\frac{p(x)}{q(x)}$ is said to be resolved into partial fractions.

2.1.1 Denominator contains non-repeated Linear factors

In the rational expression $\frac{p(x)}{q(x)}$, if $q(x)$ is the product of non-repeated linear factors of the form $(ax+b)(cx+d)$, then $\frac{p(x)}{q(x)}$ can be resolved into partial fraction of the form $\frac{A}{ax+b}+\frac{B}{cx+d}$, where A and B are constants to be determined.

For example,

$\frac{3x+7}{(x-1)(x-2)}=\frac{A}{(x-1)}+\frac{B}{(x-2)}$, the values of A and B are to be determined.

Example 2.1

Find the values of A and B if

$$\frac{1}{(x^2-1)}=\frac{A}{x-1}+\frac{B}{x+1}$$

Solution

Let $\frac{1}{(x^2-1)} = \frac{A}{x-1}+\frac{B}{x+1}$

Multiplying both sides by $(x-1)(x+1)$, we get

$1= A(x+1)+B(x-1)$ ... (1)

Put $x=1$ in (1) we get, $1 = A(2)$

$\therefore A = \frac{1}{2}$

Put $x=-1$ in (1) we get, $1=A(0)+B(-2)$

$\therefore B = -\frac{1}{2}$

Example 2.2

Resolve into partial fractions:

$$\frac{7x-1}{x^2-5x+6}$$

Solution

Write the denominator into the product of linear factors.

Here $x^2-5x+6 = (x-2)(x-3)$

Let $\frac{7x-1}{x^2-5x+6} = \frac{A}{(x-2)} + \frac{B}{(x-3)}$ ...(1)

Multiplying both the sides of (1) by $(x-2)(x-3)$, we get

$7x-1 = A(x-3)+B(x-2)$ ...(2)

Put $x = 3$ in (2),we get

$21 - 1 = B\ (1)$

$\Rightarrow \quad B = 20$

Put $x = 2$ in (2),we get

$14-1 = A(-1)$

$\Rightarrow \quad A = -13$

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